AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2017

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

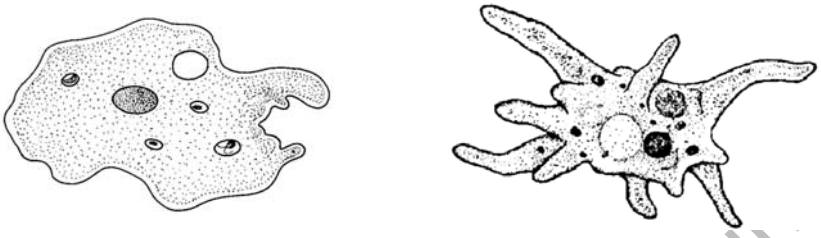
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Describe the relationship of biology with geography using an example. (2 Marks)

b. The given diagrams show two amoebae of similar size. (3 Marks)



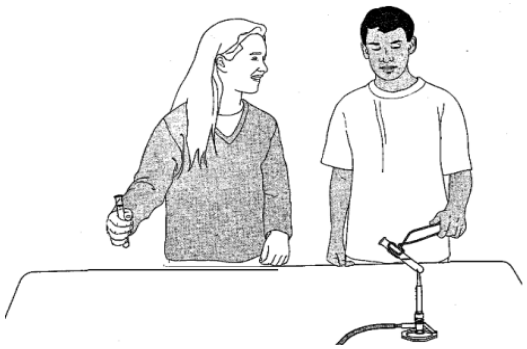
I **II**

Which of the two amoebae will have greater surface area to volume ratio? Support your answer with suitable description.

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Q.2. (Total 4 Marks)

a. The given diagram shows two students working in the biology laboratory.



Identify any TWO safety measures that the students have NOT taken while working in the laboratory. (2 Marks)

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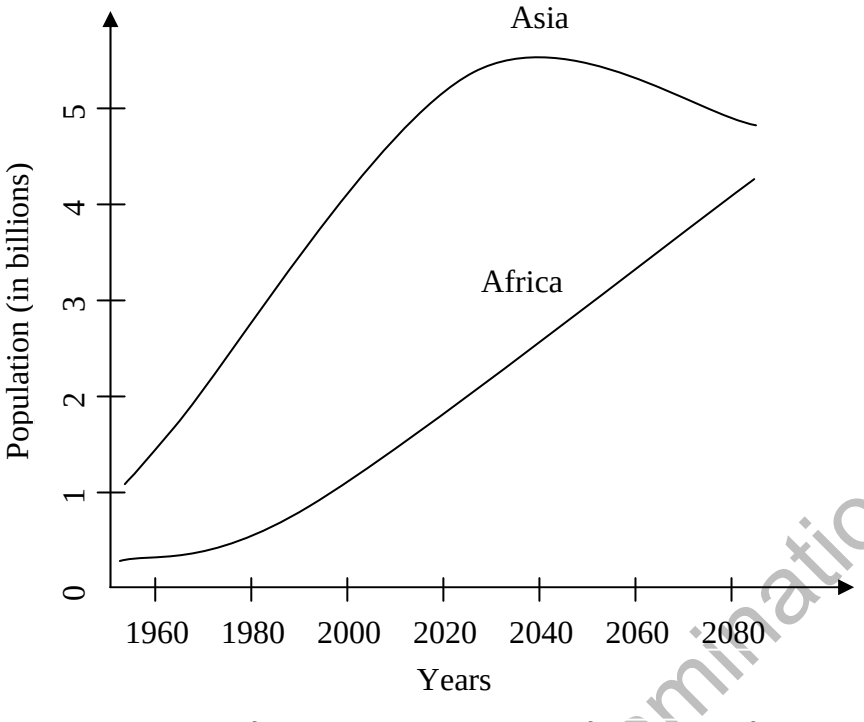
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b. The given graph shows the population growth of two different continents. (2 Marks)

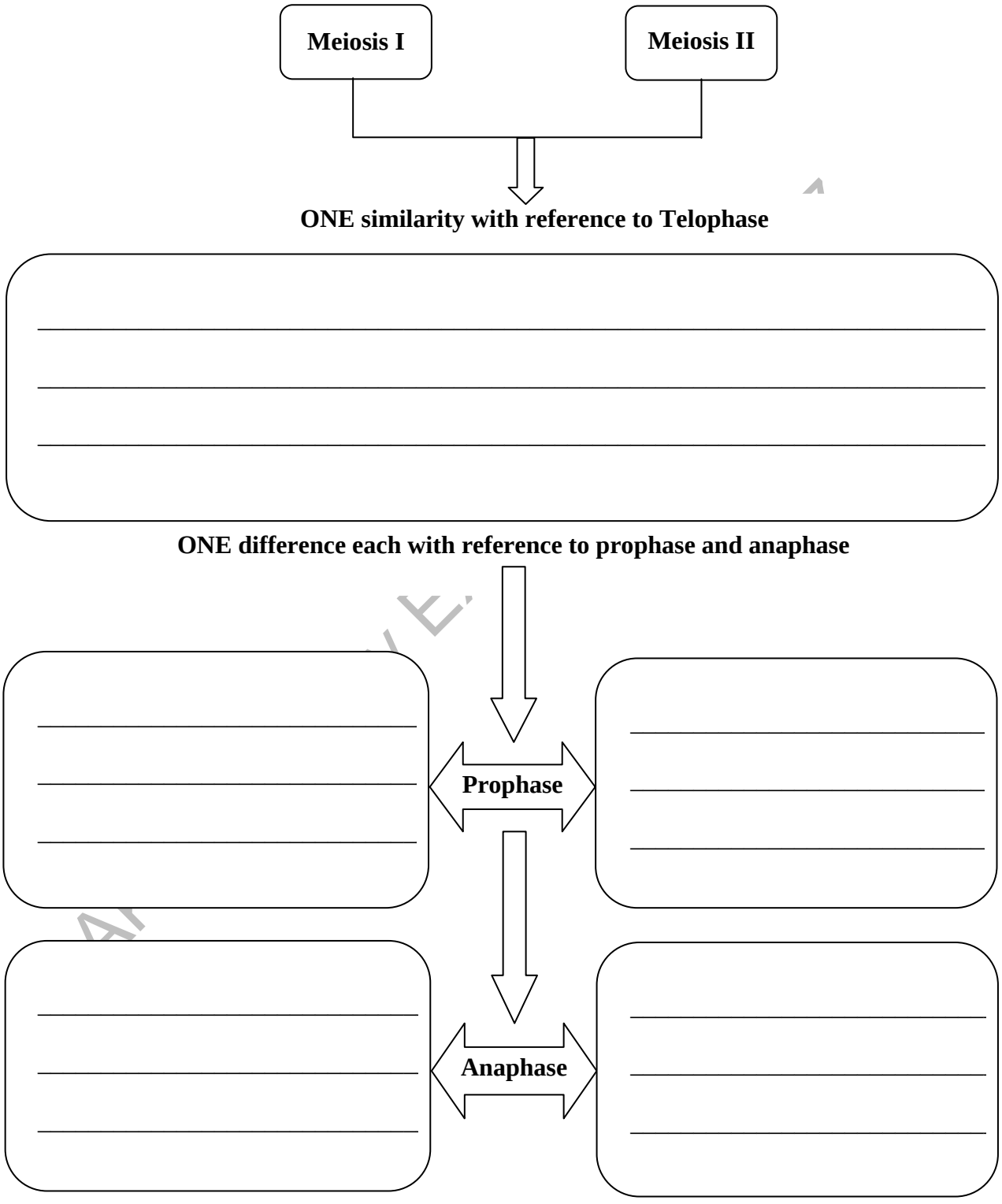


Interpret the graph referring to the population of Asia and Africa.

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Q.3. (Total 3 Marks)

Complete the given flow chart of meiosis.



Q.4. (Total 2 Marks)

Mention the general properties of enzymes for the given examples.

Example	Property
2 mg lipase is required to digest 200 g or 400 g of lipids.	
Magnesium is required for the proper functioning of ATPase.	

Q.5. (Total 3 Marks)

A student takes rest after a strenuous exercise. While resting, the lactic acid produced in the student's muscles during exercise is converted into glucose.

Outline the process of conversion of lactic acid into glucose.

Q.6. (Total 3 Marks)

State the source of the given digestive juice and name ONE enzyme present in each juice.

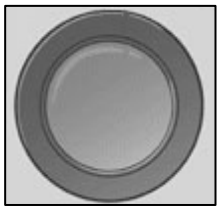
Digestive Juice	Source	Enzyme
Gastric juice	Gastric gland	
Bile juice		

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Q.7. (Total 5 Marks)

a. State any TWO functions of carbohydrates in plants. (2 Marks)

b. The given diagrams show cross-sections of arteries.



I. Normal Artery



II. Diseased Artery

i. Identify the disease shown in diagram **II**. (1 Mark)

ii. Mention any TWO factors that lead to the diseased condition of the artery as shown in diagram **II**. (2 Marks)

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Q.8. (Total 7 Marks)

EITHER

a. The given diagrams represent two human activities which cause imbalance in nature.

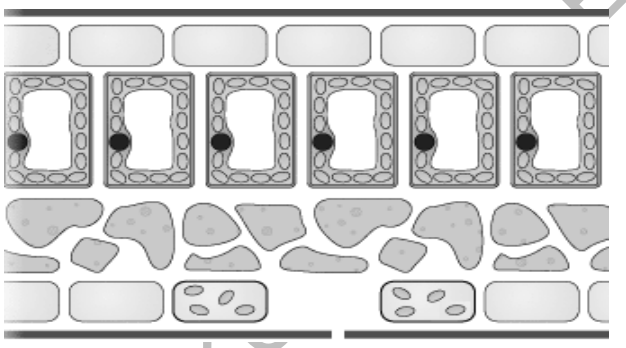




Describe the given human activities and their impact in bringing imbalance in nature with reference to biodiversity.

OR

b. The given diagram shows the cross-section of a leaf.



i. What is the function of network of veins in the leaf? (1 Mark)

ii. Describe the structural adaptation and advantages of any THREE features present in the cross-section of the leaf shown in the given diagram. (6 Marks)

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Q.9. (Total 8 Marks)

EITHER

a.

i. Mention any ONE advantage and ONE disadvantage of using organic fertilisers. (2 Marks)

ii. Explain the effects of excessive usage of inorganic fertilisers on the following: (6 Marks)

I. Global climate

II. Aquatic life

OR

b.

i. Mention any TWO structural adaptations of phloem tissue for translocation. (2 Marks)

ii. Describe the translocation of food in phloem referring to Pressure Flow Mechanism. (6 Marks)

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Please use this page for rough work

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